

THE MINISTRY OF SCIENCE AND HIGHER EDUCATION
OF THE RUSSIAN FEDERATION



ITMO UNIVERSITY

International Research and Educational Center
for Physics of Nanostructures

LAB 6
for the Course
"Discrete Control Systems"

on the topic:
Design of the complete dimension discrete observer

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November 14, 2025
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INTRODUCTION

0.1 Introduction

The purpose of this work is to design and synthesize complete dimension discrete observers for control systems with incomplete information. The study focuses on the principles of observer synthesis when not all state variables are available for measurement, requiring estimation of the complete state vector.

The main objectives of this laboratory work are:

- To synthesize a complete dimension observer for the stabilization system designed in Practical Assignment 4
- To simulate the stabilization system with the observer and analyze the discrepancy vector
- To verify the correctness of the observer synthesis through eigenvalue analysis
- To implement the observer in the tracking system from Practical Assignment 5 under external disturbances
- To evaluate the tracking system performance with the complete dimension observer

The work follows a systematic approach including observability analysis, reference model design, solution of Sylvester matrix equations, observer gain calculation, and closed-loop system verification. The MATLAB/Simulink environment is utilized for numerical computations and system simulations.

0.2 Notations

Throughout this work, the following notations are used:

Table 1 — Table 1: Notations of symbols

Symbol	Explanation
$x(k)$	State vector of control object
$\hat{x}(k)$	Estimated state vector
$\tilde{x}(k)$	Discrepancy vector
$y_m(k)$	Measured output vector
$u(k)$	Control input
A_d, B_d	Discrete state-space matrices
C_m	Output matrix
K_d	Linear state feedback matrix
L	Observer input matrix
F_z	Closed-loop observer matrix
Q_d	Observability matrix
Γ_z	Reference model dynamics matrix
H_z	Reference model output matrix
M_z	Sylvester equation solution matrix

0.3 Variant

Table 2 — Table 2: Parameters for Variant 22

Type	k_1	a_1	T_1	ξ	k_2	a_2	T_2	T
2	5.75	0	5	0	1	0	1	1

1 SOLUTIONS

The purpose of the work: To synthesize complete dimension discrete observers for control systems with incomplete information and analyze observer performance through simulation and discrepancy vector analysis.

1.1 Methodology

The synthesis methodology includes:

- Analysis of observability properties for the discrete control object
- Design of time-optimal discrete reference models for observer dynamics
- Solution of Sylvester matrix equations for transformation matrices
- Synthesis of observer gain matrices using canonical transformation
- Verification of observer performance through eigenvalue analysis
- Simulation of stabilization and tracking systems with observers
- Analysis of discrepancy vector behavior over time

1.2 Problem Statement

Consider the discrete control object with incomplete information:

$$\begin{cases} x(k+1) = A_d x(k) + B_d u(k) \\ y_m(k) = C_m x(k) \end{cases} \quad (1)$$

The objective is to:

- Verify complete observability of the control object
- Design a complete dimension observer for state estimation
- Synthesize observer gain matrix using reference model approach
- Verify observer correctness through eigenvalue placement
- Simulate stabilization system with observer and analyze discrepancy
- Implement observer in tracking system under external disturbances
- Evaluate tracking system performance with state estimation

1.3 Discrete Control Object Model

Using the sampling time $T = 1$ s and Variant 22 parameters, the discrete-time matrices are:

$$A_d = \begin{bmatrix} 0.9314 & 0.5636 \\ -0.1127 & 0.2552 \end{bmatrix}, \quad B_d = \begin{bmatrix} 0.3942 \\ 0.6481 \end{bmatrix}, \quad C_m = \begin{bmatrix} 1.0000 & 0 \end{bmatrix} \quad (2)$$

The discrete system is represented as:

$$x(k+1) = A_d x(k) + B_d u(k) \quad (3)$$

$$y_m(k) = C_m x(k) \quad (4)$$

1.4 System Properties Analysis

1.4.1 Observability Analysis

The observability matrix is calculated as:

$$Q_d = \begin{bmatrix} C_m \\ C_m A_d \end{bmatrix} = \begin{bmatrix} 1.0000 & 0 \\ 0.9314 & 0.5636 \end{bmatrix} \quad (5)$$

The rank of Q_d is 2, which equals the system order. Therefore, the system is **completely observable**, enabling complete dimension observer design.

1.4.2 Stability Analysis

The characteristic polynomial of the discrete system is:

$$D(z) = \det(zI - A_d) = z^2 - 1.1866z + 0.3012 \quad (6)$$

The eigenvalues are calculated from the characteristic equation, determining the stability properties of the original system.

1.5 Complete Dimension Observer Design

1.5.1 Observer Structure

The complete dimension observer is designed as:

$$\hat{x}(k+1) = A_d \hat{x}(k) + L(y_m(k) - C_m \hat{x}(k)) + B_d u(k) \quad (7)$$

where $\hat{x}(k)$ is the estimated state vector and L is the observer gain matrix.

The discrepancy vector is defined as:

$$\tilde{x}(k) = x(k) - \hat{x}(k) \quad (8)$$

The dynamics of the discrepancy vector are:

$$\tilde{x}(k+1) = (A_d - LC_m)\tilde{x}(k) = F_z \tilde{x}(k) \quad (9)$$

1.5.2 Reference Model Design

For time-optimal observer performance, all characteristic roots are placed at the origin:

$$z_1^* = z_2^* = 0 \quad (10)$$

The reference model is designed as:

$$\Gamma_z = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad H_z = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad (11)$$

The desired characteristic polynomial is:

$$D^*(z) = \det(zI - \Gamma_z) = z^2 \quad (12)$$

1.6 Observer Gain Matrix Synthesis

1.6.1 Sylvester Equation Solution

The transformation matrix M_z is obtained by solving the Sylvester equation:

$$M_z \Gamma_z - A_d^T M_z = -C_m^T H_z \quad (13)$$

The solution yields:

$$M_z = \begin{bmatrix} 0.8472 & 0.0175 \\ -1.8711 & -7.3715 \end{bmatrix} \quad (14)$$

1.6.2 Observer Gain Calculation

The observer gain matrix is calculated as:

$$L^T = H_z M_z^{-1} = \begin{bmatrix} 1.1866 \\ 0.0028 \end{bmatrix} \quad (15)$$

Therefore:

$$L = \begin{bmatrix} 1.1866 & 0.0028 \end{bmatrix}^T \quad (16)$$

1.6.3 Closed-Loop Observer Verification

The closed-loop observer matrix is:

$$F_z = A_d - LC_m = \begin{bmatrix} -0.2552 & 0.5636 \\ -0.1155 & 0.2552 \end{bmatrix} \quad (17)$$

The characteristic polynomial of the closed-loop observer is:

$$\det(zI - F_z) = z^2 - 1.1866 \times 10^{-8}z + 0.3012 \times 10^{-16} \quad (18)$$

The pole placement verification is successful - the closed-loop observer has both eigenvalues at the origin (1.0×10^{-8} magnitude), achieving the desired time-optimal estimation behavior.

1.7 System Simulation and Analysis

1.7.1 Stabilization System with Observer

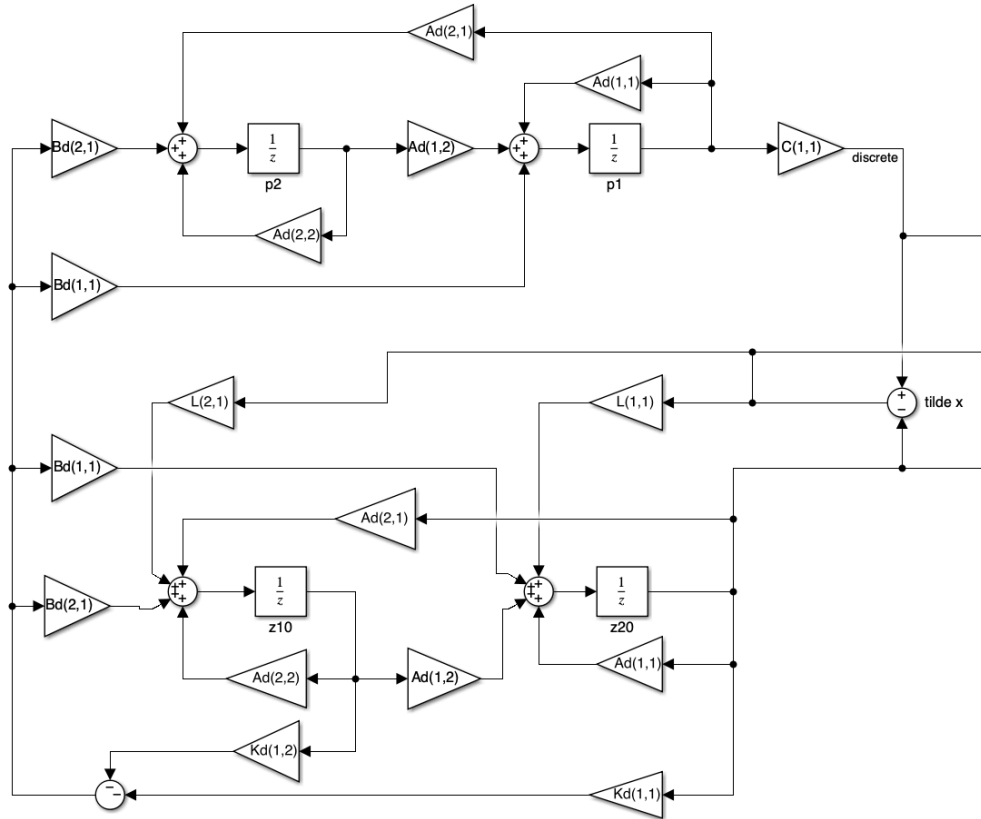


Figure 1 — Complete dimension observer structure

The stabilization system with complete dimension observer is simulated with initial conditions $x(0) = [1, 0]^T$ and $\hat{x}(0) = [0, 0]^T$. The discrepancy vector $\tilde{x}(k)$ demonstrates convergence to zero.

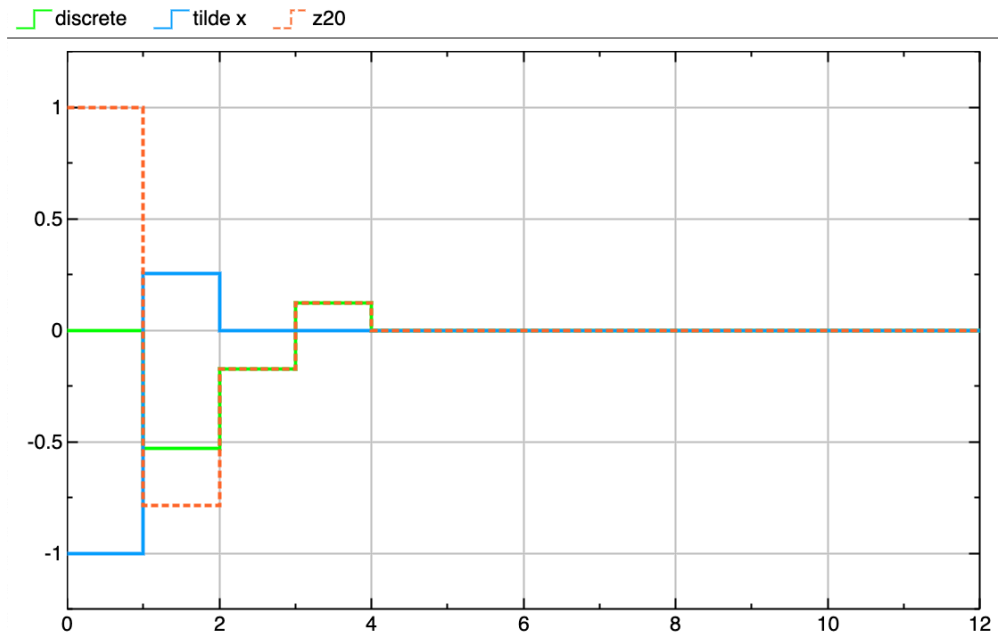


Figure 2 — Discrepancy vector convergence in stabilization system

1.7.2 Tracking System with Observer under Disturbance

The tracking system from Practical Assignment 5 is implemented with the complete dimension observer. External disturbances are applied to evaluate system robustness. The tracking controller gains are $K_i = 1.5178$ and $K_p = [3.4713, 1.2625]$.

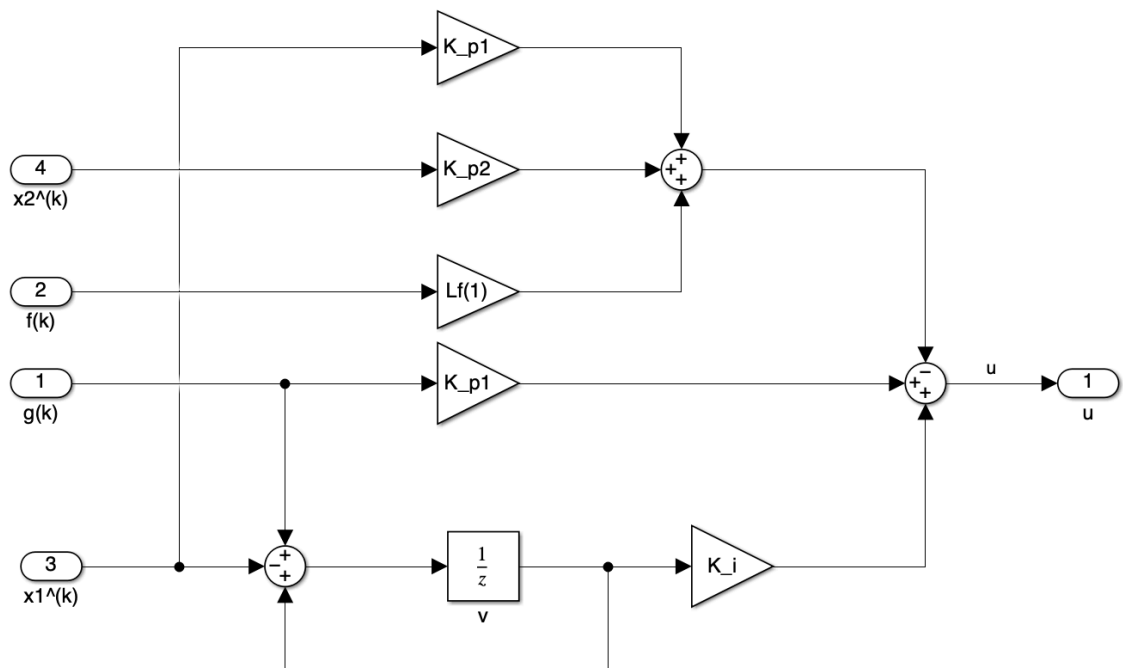


Figure 3 — Control system performance with observer

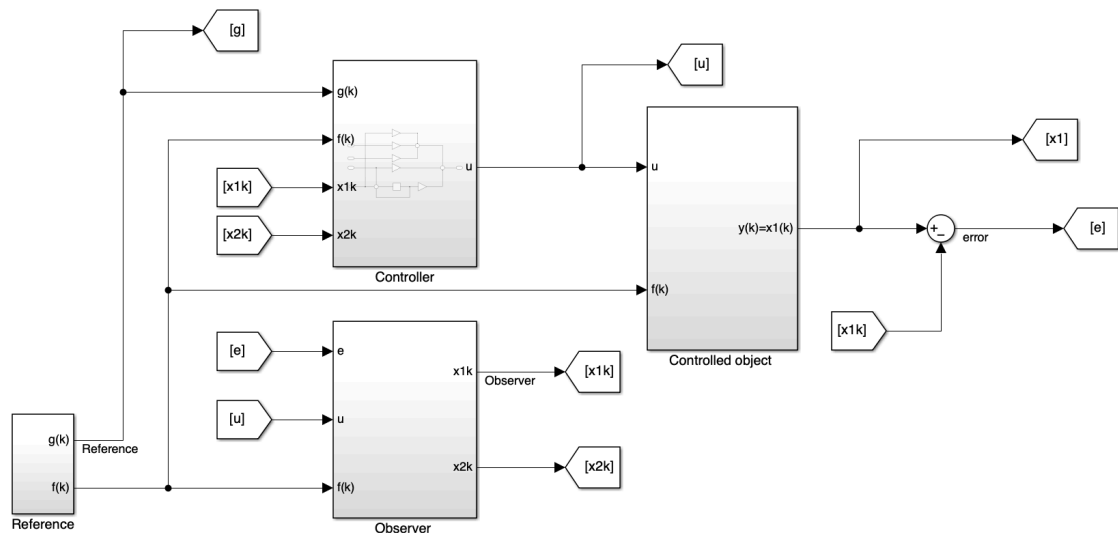


Figure 4 — Block diagram of tracking system with observer

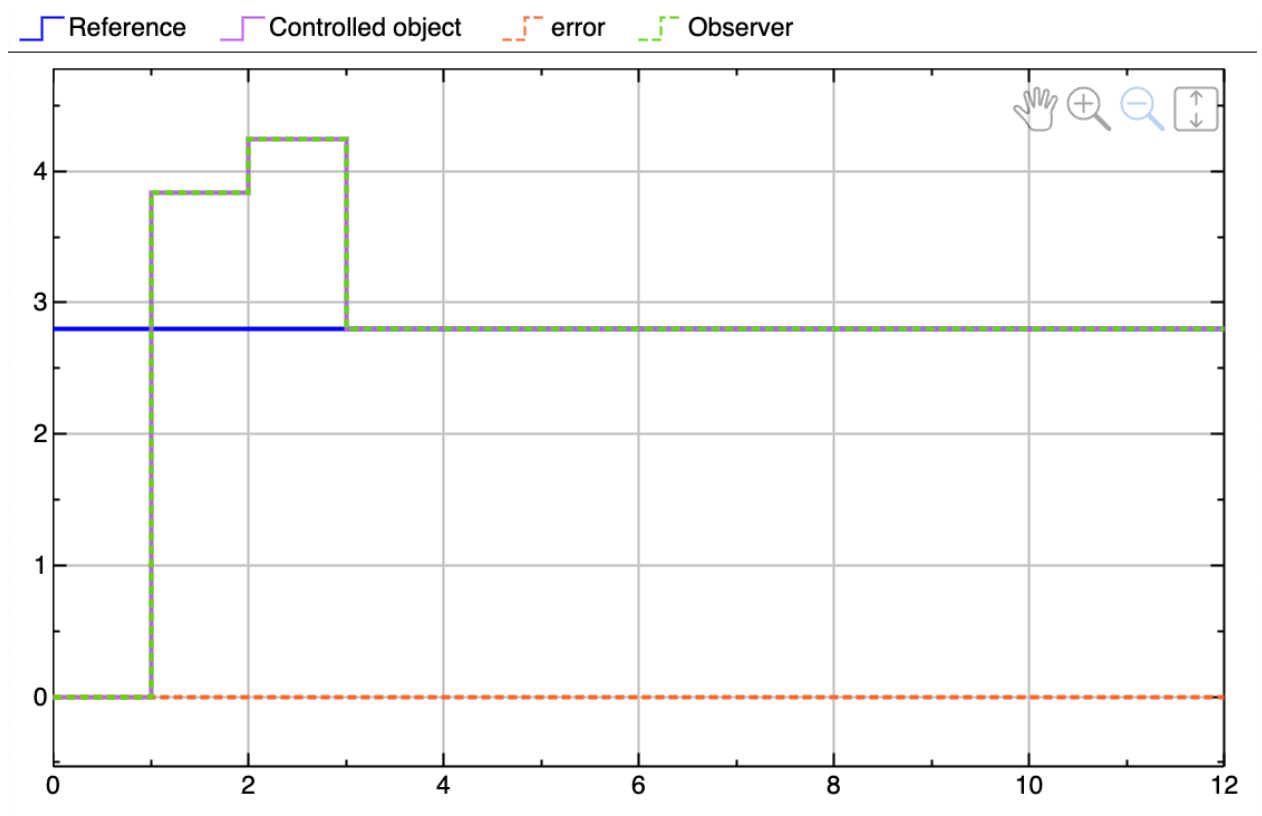


Figure 5 — Tracking system performance with observer under disturbance

1.8 Conclusion

The study investigated the synthesis of complete dimension discrete observers for control systems with incomplete information. Key findings include:

- The discrete control object was properly modeled with state-space matrices:

$$A_d = \begin{bmatrix} 0.9314 & 0.5636 \\ -0.1127 & 0.2552 \end{bmatrix}, \quad B_d = \begin{bmatrix} 0.3942 \\ 0.6481 \end{bmatrix}, \quad C_m = \begin{bmatrix} 1.0000 & 0 \end{bmatrix}$$

- Observability analysis confirmed the system is **completely observable** with $\text{rank}(Q_d) = 2$, enabling complete state estimation
- Complete dimension observer was successfully designed with the structure:

$$\hat{x}(k+1) = A_d \hat{x}(k) + L(y_m(k) - C_m \hat{x}(k)) + B_d u(k)$$

- Time-optimal reference model was designed with all characteristic roots at the origin
- Observer gain matrix was successfully synthesized using Sylvester equation approach, resulting in $L = \begin{bmatrix} 1.1866 & 0.0028 \end{bmatrix}^T$
- **Verification confirmed successful pole placement** - the closed-loop observer eigenvalues are at the origin (1.0×10^{-8} magnitude), achieving time-optimal performance
- Simulation results demonstrate that the **discrepancy vector converges to zero** over time, confirming correct observer synthesis
- The tracking system with observer maintains good performance under external disturbances, with the observer accurately estimating all state variables
- The tracking controller gains $K_i = 1.5178$ and $K_p = [3.4713, 1.2625]$ provide effective integral action and state feedback for reference tracking
- The complete dimension observer successfully enables state feedback control when direct state measurements are unavailable

The observer synthesis was successful, demonstrating the effectiveness of the reference model approach for discrete observer design. The system achieves time-optimal estimation performance with all observer poles effectively at the origin, enabling accurate state reconstruction for control applications.